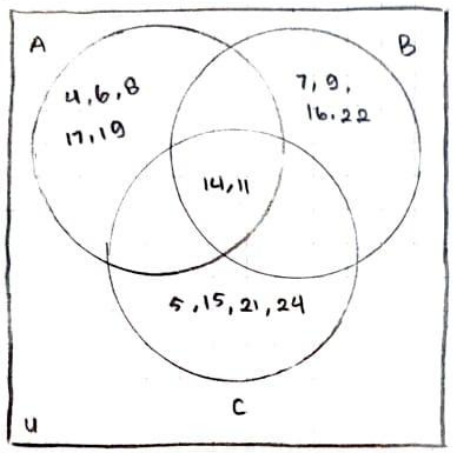
③ A = { 4, 6, 8, 11, 14, 17, 19 } → media sosial pelangan
B = { 7, 9, 11, 14, 16, 22 } → referal
C = { 5, 11, 14, 15, 21, 24 } → loyality program

U = { 4, 5, 6, 7, 8, ... , 24 }

a



$A \cup B \cup C$ = {4, 5, 6, 7, 8, 9, 11, 14, 15, 16, 17, 19, 21, 22, 24}

$A \cap B \cap C$ = {14, 11}

$(A \cup B) \cap C'$ =
- $A \cup B$ = {4, 6, 7, 8, 9, 11, 14, 16, 17, 19, 22}
- C' = {4, 6, 7, 8, 9, 10, 12, 13, 16, 17, 18, 19, 20, 22, 23}

= {4, 6, 7, 8, 9, 16, 17, 19, 22}

$A \oplus B$ = $(A \cup B) - (A \cap B)$
- $A \cup B$ = {4, 6, 7, 8, 9, 11, 14, 16, 17, 19, 22}
- $A \cap B$ = {11, 14}

= {4, 6, 7, 8, 9, 16, 17, 19, 22}

b MATLAB

c $F(x) = 5x + 4$

$g(x) = \frac{x^2}{5}$

- $(F \circ g)(x) = \cancel{5}\left(\frac{x^2}{\cancel{5}}\right) + 4 = x^2 + 4$

$(g \circ F)(x) = \frac{(5x+4)^2}{5} = \frac{25x^2 + 40x + 16}{5} = 5x^2 + 8x + \frac{16}{5}$

- cek komposisi komutatif

$x = 6$

- $F(g(6))$

$g(6) = \frac{6^2}{5} = 72$

$F(7.2)$:

$F(7.2) = 5(7.2) + 4 = 40$

- $g(F(6))$ =

$F(6) = 5 \cdot 6 + 4 = 34$

$g(28) = \frac{34^2}{5} = 231.2$

karena $F(g(6)) \neq g(F(6))$ maka komposisinya tidak komutatif

d. Pendapatan total = Rp. 404.000

$$F^{-1}(x) = \frac{x-(a+b)}{k}$$

$$= \frac{x-4}{4}$$

$$F^{-1}(404) = \frac{404-4}{4} = 100 \text{ unit} \ldots$$

Jadi volume pembelian x yang menghasilkan pendapatan Rp. 404.000 adalah 100 unit..

e. Fungsi $F(x) = 4x + 4$ adalah bijektif karna memenuhi syarat injektif & surjektif

Fungsi $g(x) = \frac{x^2}{5} + 0$ adalah surjektif pada interval $[a, \infty]$